Simulating Entropy Evolution

Junyu Hu, Calvin Stubbins

Department of Physics & Astronomy, Franklin & Marshall College

Introduction

This study, starting from the second law of thermodynamics, aims to precisely determine the formula for entropy. The investigation is conducted using simulations based on two primary models

1. Ideal Gas Model: This model utilizes the Maxwell distribution to generate particles with random velocities.
2. Random Walk Model: The random walk model simulation will support and refine the assumptions made. [1]

Basic Formula Derivation of Entropy

In an isolated system, where the initial entropy is $S(0)$ at time t=0, the rate of change of the entropy is given as [1]

$$\frac{dS}{dt} = f(S - S_f) \tag{1}$$

- S_f represents the entropy when the system reaches thermal equilibrium. Performing a second-order Taylor expansion of the above formula gives

$$\frac{dS}{dt} = f(S - S_f) = \frac{df}{dS}(S - S_f) + \frac{1}{2!}\frac{d^2f}{dS^2}(S - S_f)^2$$

or

$$\frac{dS}{dt} = a_1(S - S_f) + a_2(S - S_f)^2$$

In terms of the roots of the RHS,

$$\frac{dS}{dt} = a_2(S - S_f)(S - S_i) \tag{2}$$

- S_i is the initial entropy of the system and $S_i = S_f - \frac{a_1}{a_2}$
- For any system: $S_i \le S(0) < S_f$

Integrating to find the formula *S(t)*

$$\int_{S(0)}^{S(t)} \frac{1}{(S - S_f)(S - S_i)} = \int_0^t a_2 t\, dt$$

Finally, the function S(t) is

$$S(t) = \frac{S_f - S_i \frac{S(0) - S_f}{S(0) - S_i} e^{a_2(S_f - S_i)t}}{1 - \frac{S(0) - S_f}{S(0) - S_i} e^{a_2(S_f - S_i)t}} \tag{3}$$

Similarly, higher-order entropy formulas can be derived.

The derivative function in $f(s)$ is shown in Figure.1

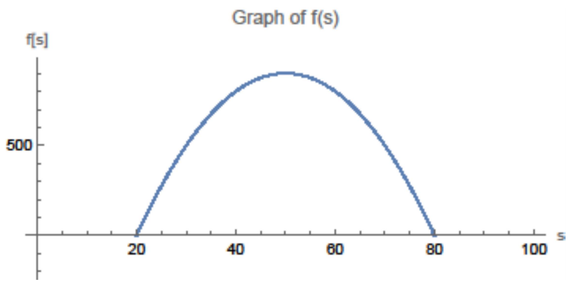


Fig. 1. Graph of function $f(S)$ with constant a and roots at s=20, s=80.

Entropy Calculation

Gibbs Formula

The calculation of entropy in this study follows Gibbs' method, which divides the system into multiple cells. The total entropy of the system is the sum of the entropy of each cell is

$$S_{tot} = \sum_{i=1}^{n} \mathcal{S}_i$$

- n is the number of cells

For each cell, $\mathcal{S}_i$ is calculated using Gibbs free energy,

$$\mathcal{S} = -k_B\, P\, lnP$$

- k_B is the Boltzmann constant
- P is the probability of a particle being in a certain cell, given by

$$P = \frac{N}{N_{tot}}$$

- N is the number of particles in a cell
- N_{tot} is the total number of particles in this system

Model Simulation

This research utilizes two models to simulate the movement of particles in an isolated system. Both models share the same initial setting for the particle positions.

- The simulation starts with a Square box with length $\boldsymbol{L}$
- The initial positions of all particles are randomly generated within a smaller box of side length L_0 at the center of the larger square box
- As the simulation starts, the smaller box disappears
- The exact movement of each particle is determined by the respective model.

Ideal Gas

In this model, the velocities of the particles are generated according to the Maxwell-Boltzmann distribution.[2]

After generating velocities in the three dimensions (v_x, v_y, v_z), the maximum velocities for each dimension $(v_{xmax}, v_{ymax}, v_{zmax})$ are determined.

For the x-dimension at time t, the phase-space particle coordinates are shown in Figure 2.

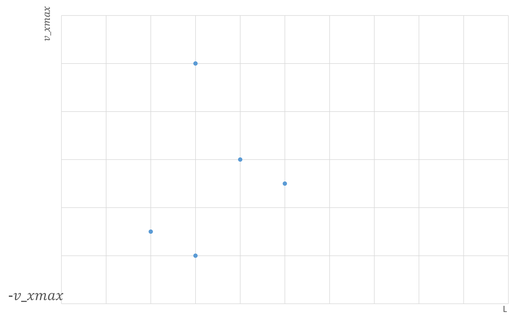


Fig. 2. Division of cells in phase-space

The coordinate system divides the 6-D phase-space into cells. Entropy is calculated for each cell using Gibbs free energy.

Random Walk

This model simulates particles moving in random directions. The key constant for this model is $\vec{v}$, the velocity of each particle at every time step t.

The cells are arranged as shown in Figure 3. The total system entropy is the sum of the entropy from all cells.

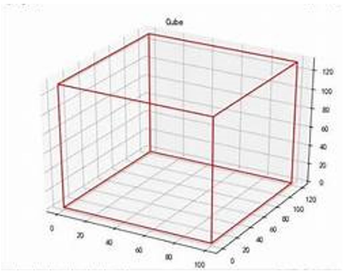


Fig. 3. Division of cells for random walk model

Statistical Methods and Data Fitting

For each model, data from 50 experiments is collected. By using a large sample size, the average and standard deviation naturally account for any outliers, ensuring more reliable and robust results. These values are then used for analysis and graphing.[3]

Curve Fit

Simulated data from 50 experiments was used to fit the function and find the parameters a_2, S_i, S_f and $S(0)$

Results

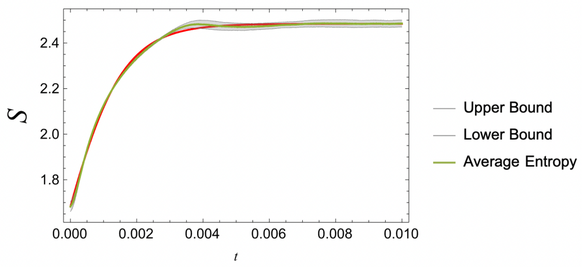


Fig. 4 The scatter plot and curve fit of ideal gas model S vs t

	Estimate	Standard Error	t-Statistic	P-Value
a1	−1104.82	8.25275	−133.873	0.
a2	−566.348	15.2669	−37.0964	5.11551×10^{-190}
Sf	2.48412	0.000374036	6641.38	0.

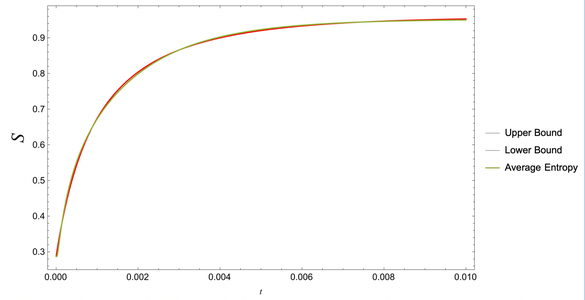


Fig. 5. *S* vs *t* scatter plot and curve fit of random walk model with 500k particles

	Estimate	Standard Error	t-Statistic	P-Value
a1	380.112	3.87736	98.0337	0.
a2	1103.44	8.54482	129.136	0.
Sf	1.30253	0.00576719	225.852	0.

Conclusions

- The model in Eq. (1) of the rate of change of the entropy in terms of a function of the entropy leads to a valid model of the time dependence of the entropy
- The second order approximation to $f(S)$ in Eq. (2) leads to an expression of $S(t)$ in Eq. (3) that is very good model of the time dependence of the entropy for two very different physical systems

Future Studies

An analytical model of entropy for Brownian motion is proposed in reference [4].

$$S(t) = -\left(\frac{1}{2} - \frac{e^{-t}}{2}\right)\ln\left(\frac{1}{2} - \frac{e^{-t}}{2}\right) - \left(\frac{1}{2} + \frac{e^{-t}}{2}\right)\ln\left(\frac{1}{2} + \frac{e^{-t}}{2}\right)$$

However, the expression in Eq. (2) for $f(S)$ is not general enough to properly fit this model. This model is also singular at $t = 0$. We expect that a Laurent series of the form

$$f(S) = \sum_{i=-m}^{n} a_i S^i$$

should be used in Eq.(2)

References

[1] Jean-Bernard, B., " The meanings of entropy," Department of Physics, Lab/UFR High Energy Physics, Feb 14, 2005.

[2] Subhadip, C., Abhishek, D., Anupam, K., "Boltzmann's Entropy During Free Expansion of an Interacting Gas," Journal of Statistical Physics (2023) , March 15, 2023

[3] Harry, L., Bell, T., "An Approach to Equilibrium", PHYSICAL REVIEW, January 1, 1958.

[4] Mesfin A, "Free energy and entropy production rate for a Brownian particle that walks on overdamped medium", PHYSICAL REVIEW E94 (2016), September 9, 2016.

www.PosterPresentations.com